

PlayfulOutings

Picnic Games & Getaways

eProject Documentation Report

Semester 1 eProject — HTML5 Web Development

Aptech Computer Education

Submitted by:

Chukwu Nathan Emeka

Daniel Fortune Rohi

Chukwu Stanley Ebuka

Daniel Chinonso Favour

Date: March 2026

Location: Abuja, Nigeria

Making every outing memorable • Abuja, Nigeria • March 2026

Table of Contents

1. Acknowledgements

2. eProject Synopsis

3. Problem Definition

4. eProject Analysis

5. eProject Design

— DFD — Context Diagram

— User Navigation Flowchart

— System Internal Flowchart

— Process Diagram

6. Design Specifications

7. Screenshots Description

8. Source Code with Comments

9. User Guide

10. Developer's Guide

— Module Descriptions

1. Acknowledgements

We would like to express our sincere gratitude to everyone who contributed to the successful completion of this eProject. First and foremost, we thank **Aptech Computer Education** and the **eProjects Team** for providing this structured learning opportunity and for guiding us through the practical application of web development concepts.

We are grateful to our course instructors and faculty members whose lessons on HTML5, CSS3, and JavaScript formed the foundation of everything we built in this project. Their support during lab sessions was invaluable.

Special thanks to our group members — **Chukwu Nathan Emeka**, **Daniel Fortune Rohi**, **Chukwu Stanley Ebuka**, and **Daniel Chinonso Favour** — for their dedication, teamwork, and commitment throughout the development process.

Finally, we acknowledge the open-source community, YouTube content creators whose tutorial videos are embedded in our platform, and Google Maps for the mapping functionality that enhances our Contact section.

2. eProject Synopsis

Project Title	PlayfulOutings — Picnic Games & Getaways
Project Type	Single-Page Application (SPA) Website
Technology Stack	HTML5, CSS3, Vanilla JavaScript
Team Members	Chukwu Nathan Emeka, Daniel Fortune Rohi, Chukwu Stanley Ebuka, Daniel Chinonso Favour
Date	March 2026
Location	Abuja, Nigeria
Institution	Aptech Computer Education
Course	Semester 1 — HTML5 Web Development

PlayfulOutings is a fully client-side, responsive Single-Page Application designed to help families, couples, youth groups, and friends in Abuja discover and organize enjoyable picnic games and short outdoor getaways. The platform features 24 curated games across 6 categories, complete with materials, time requirements, step-by-step instructions, embedded tutorial videos, and downloadable Word document guides. It also includes Abuja-specific itinerary suggestions, geolocation integration, a live ticker, and a persistent visitor counter.

3. Problem Definition

In urban cities like Abuja, many families, couples, youth groups, and friends find it challenging to organize enjoyable picnics and short outdoor outings. Common issues include:

Lack of ready-to-use games: No single platform provided age-appropriate games with clear instructions, required materials, and estimated playtime in one place.

Repetitive gatherings: Without structured activities, picnics become boring and unmemorable, reducing the quality of social bonding time.

No localized getaway plans: Existing resources did not cater to Abuja-specific popular outdoor locations like Jabi Lake, Millennium Park, or Usuma Dam.

Difficulty organizing groups: Organizing games for mixed groups (kids, adults, males, females) required searching multiple sources with no unified solution.

Solution: PlayfulOutings solves these problems by providing a single, easy-to-use web platform that offers curated indoor and outdoor picnic games categorized for different groups, step-by-step game guides, Abuja-specific itinerary suggestions, and interactive features to enhance user engagement — all accessible from any device with a web browser.

4. eProject Analysis

Functional Requirements

- Display a categorized library of 24 picnic games (Indoor, Outdoor, Kids, Males, Females, Family).
- Provide Materials Needed, Time Required, and How-to-Play instructions for each game.
- Embed YouTube demo videos for every game with lazy loading for performance.
- Offer downloadable Word document (.docx) instruction sheets for each game.
- Include an Itinerary section with Abuja-specific short getaway suggestions.
- Display an About Us and Contact Us section with a Google Maps embed.
- Implement a category filter (sub-navigation) that shows only the selected game group.
- Show a persistent visitor counter using browser localStorage.
- Display a live scrolling ticker showing current date, time, and user location.
- Detect and display user geolocation in the Contact section and ticker.
- Provide a dropdown menu under "Games" in the header for quick category access.

Non-Functional Requirements

Requirement	Description
Performance	Fast load time via lazy-loaded videos; no backend server required.
Responsiveness	Fully mobile-friendly layout using CSS Grid, Flexbox, and media queries.
Compatibility	Works across Chrome, Firefox, Edge, and Safari browsers.
Usability	Intuitive navigation with smooth scrolling, active link highlighting, and clear visual hierarchy.
Maintainability	Clean, well-commented code organized into three files (HTML, CSS, JS).

5. eProject Design

Data Flow Diagram (DFD) — Context Diagram

The Level 0 DFD shows the PlayfulOutings system as a single process and its interaction with all external entities: the User (browser), localStorage (visitor data), YouTube API (video embeds), and the Geolocation API (location data).

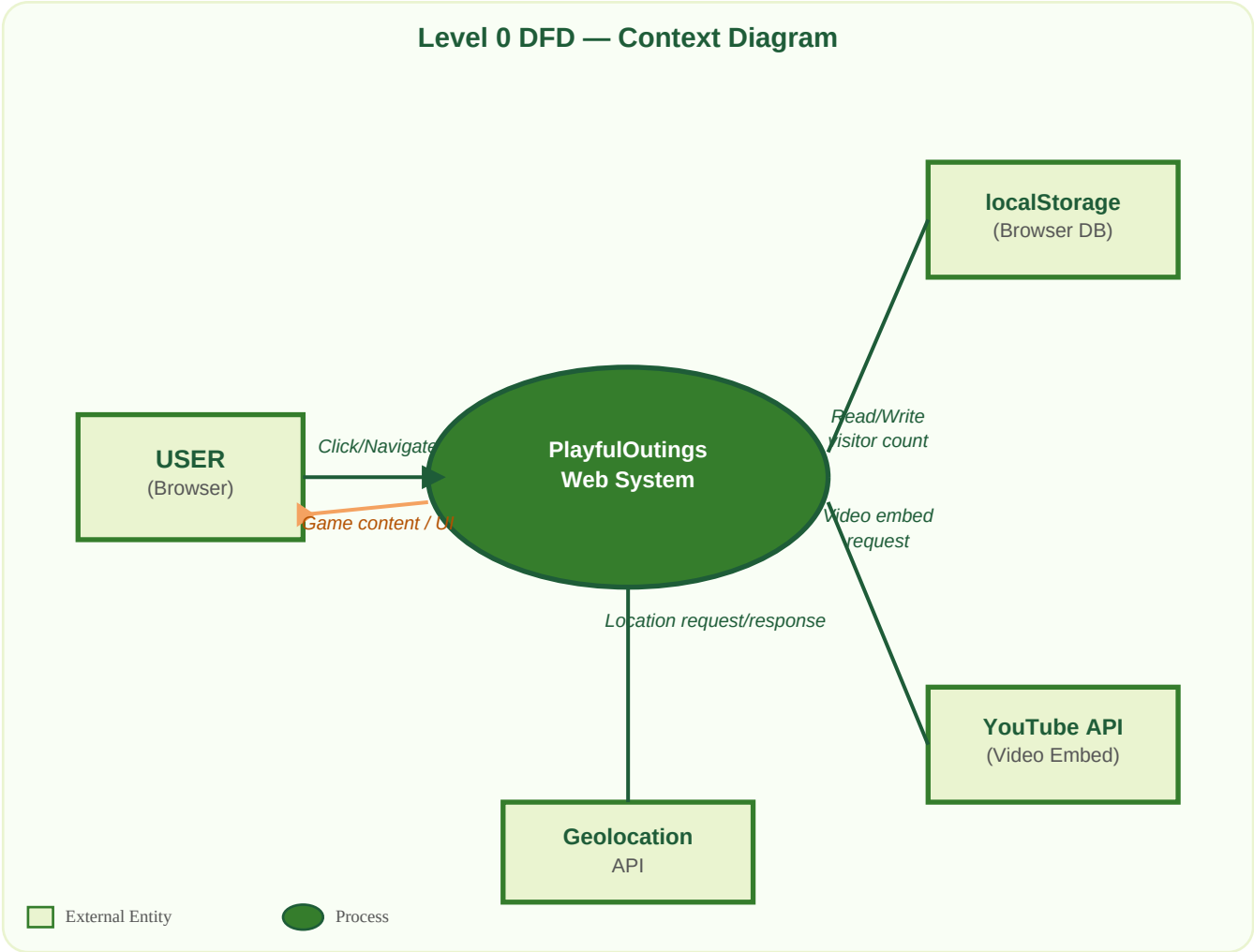


Figure 1: Level 0 Context DFD for PlayfulOutings

DFD Symbol Key

Rectangle = External Entity (User, APIs, localStorage)

Ellipse = Process (the PlayfulOutings system)

Arrow = Data Flow direction

Flowchart 1 — User Navigation Flow

This flowchart shows how a user interacts with the website from opening it to consuming content. Standard flowchart symbols are used throughout: ovals for Start/End, rectangles for processes, and diamonds for decisions.

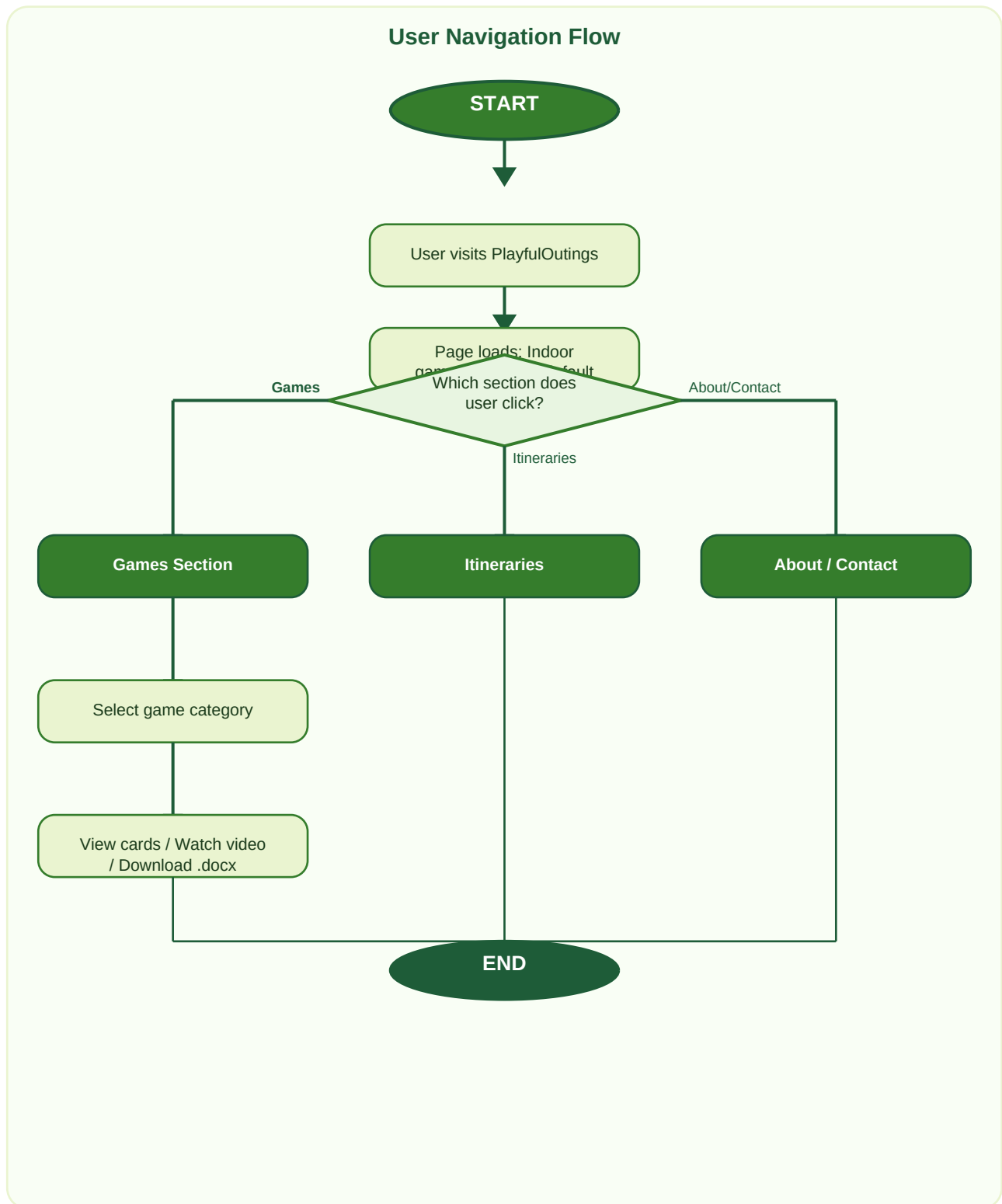


Figure 2: User Navigation Flowchart

Flowchart Symbol Key

Oval (green) = Start / End terminal

Rectangle (light green) = Process / Action step

Diamond = Decision point (branch)

Arrow = Flow of control

Flowchart 2 — System Internal Flow

This flowchart shows what happens internally when the page loads — how JavaScript initialises the visitor counter, hides all game categories and shows Indoor by default, requests geolocation, and starts the live ticker.

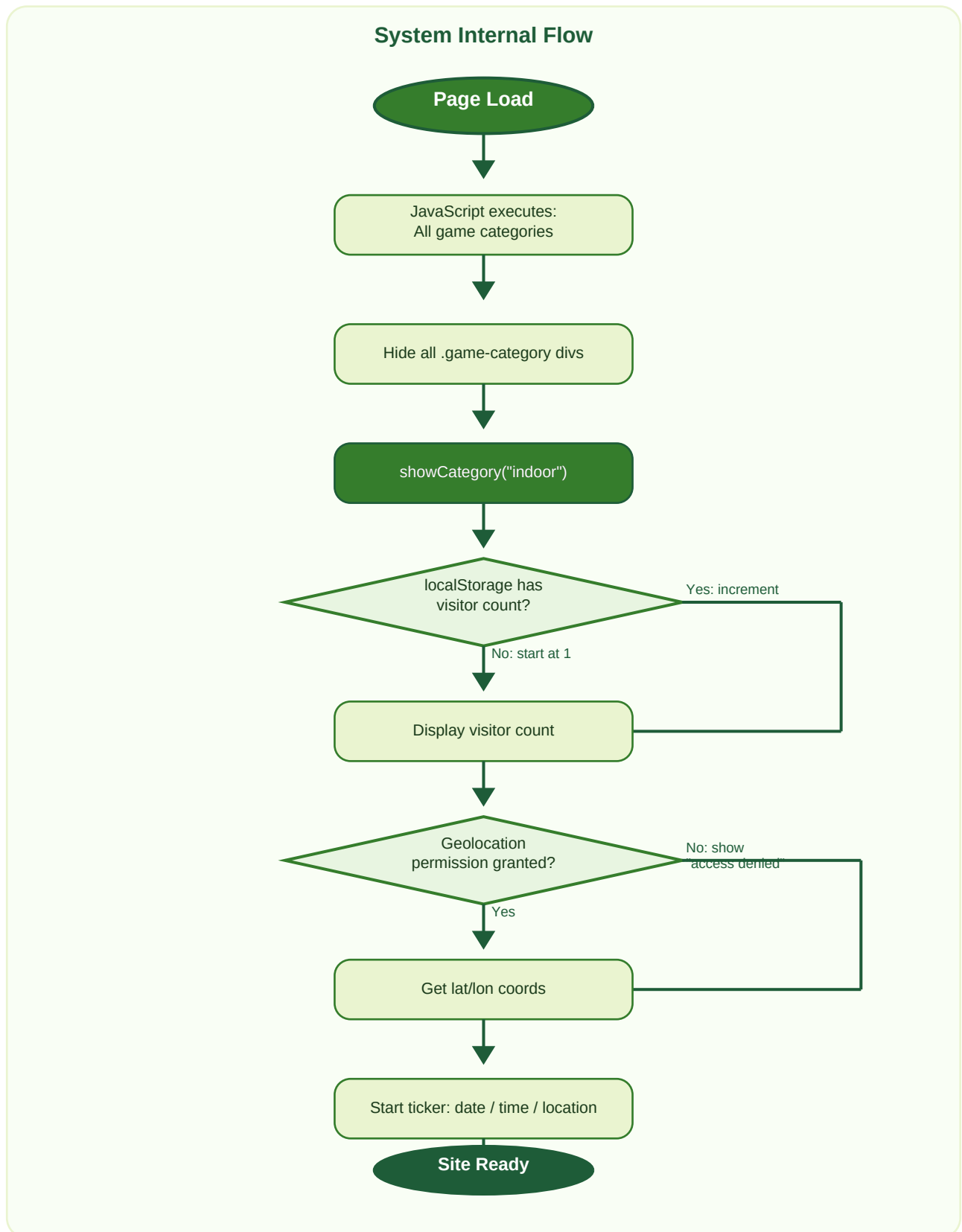


Figure 3: System Internal Flow on Page Load

Process Diagram — Game Category Filter

This diagram illustrates the process flow when a user selects a game category either from the sub-navigation bar or the header dropdown menu.

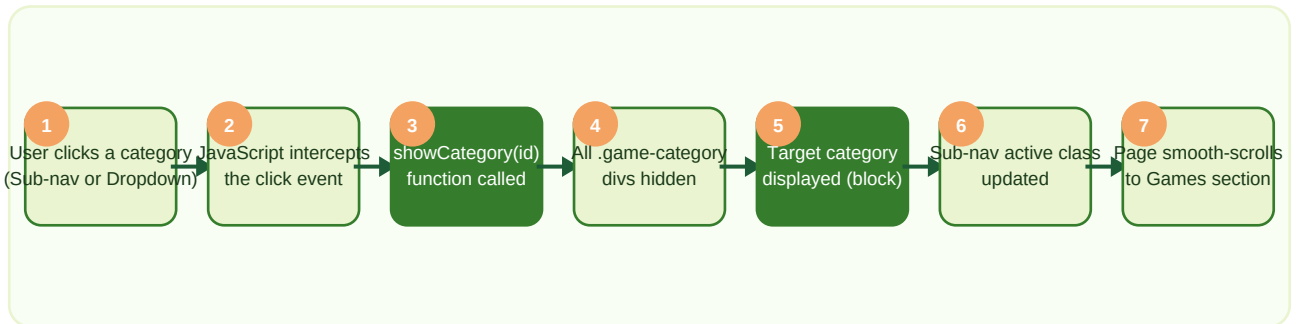


Figure 4: Game Category Filter Process Diagram

6. Design Specifications

Technology Stack

Technology	Purpose	Details
HTML5	Structure	Semantic tags, sections, iframes, geolocation API
CSS3	Styling	Glassmorphism, CSS Variables, Flexbox, Grid, Animations
JavaScript (ES6)	Interactivity	DOM manipulation, localStorage, Geolocation, Event listeners
Google Maps Embed	Location	Millennium Park map in Contact section
YouTube Embed API	Video	Lazy-loaded game tutorial videos
localStorage	Data Store	Persistent visitor count across sessions
Google Fonts	Typography	Poppins (body), Press Start 2P (hero title)

Color Scheme & Visual Design

CSS Variable	Hex Value	Usage
--primary	#357d2c	Buttons, labels, accents, card headers
--primary-dark	#1e5c38	Hover states, darker accents
--accent	#f4a261	Call-to-action highlights
--glass	rgba(255,255,255,0.12)	Glassmorphism navigation and sections
--glass-border	rgba(255,255,255,0.25)	Glass element borders
--text	#ffffff	Primary text on dark backgrounds

7. Screenshots Description

The following describes each key screen of the PlayfulOutings website. Screenshots should be captured from the live site and inserted here for the final submission.

Screen 1 — Hero / Home Section

The landing view features the "Welcome to Playful Outings" title in Press Start 2P font over a full-screen looping outdoor picnic video. A glass-effect navigation bar sits at the top with the logo, nav links (including a Games dropdown), and visitor counter. An "Explore Games" CTA button is centered below the title.

[Screenshot: Screen 1 — Hero / Home Section]

Screen 2 — Games Section with Filter

Below the hero, the Games section displays the sub-navigation pills (Indoor, Outdoor, Kids, Males, Females, Family). By default, Indoor games are shown. Each game card has a colored header bar, materials, time, how-to-play info, a YouTube thumbnail with red play button, and a green "Download Instructions (.docx)" button.

[Screenshot: Screen 2 — Games Section with Filter]

Screen 3 — Category Filter Active (e.g., Outdoor)

When a user clicks "Outdoor" in the sub-nav, only the 4 Outdoor game cards are visible. The active sub-nav pill is highlighted with a brighter background and green border. All other categories are hidden.

[Screenshot: Screen 3 — Category Filter Active (e.g., Outdoor)]

Screen 4 — Itineraries Section

A glassmorphism card section listing three Abuja-specific getaway itineraries: Jabi Lake Day, Millennium Park Picnic, and Usuma Dam Outing, with suggested spots listed below.

[Screenshot: Screen 4 — Itineraries Section]

Screen 5 — Contact Section

Displays contact email, phone, and a live geolocation readout. Below that is an embedded Google Maps iframe showing Millennium Park, Abuja. A simple contact form with name, message fields and a Send button rounds out the section.

[Screenshot: Screen 5 — Contact Section]

Screen 6 — Mobile View

On screens under 768px, the header stacks vertically, nav links become full-width, game cards go single-column, and the sub-nav wraps in a flex row. The ticker remains pinned at the bottom.

[Screenshot: Screen 6 — Mobile View]

8. Source Code with Comments

index.html — Key Structure

```

<!-- BACKGROUND VIDEO: fixed full-screen looping picnic video -->
<video id="bg-video" autoplay muted loop playsinline>
<source src="videos/bg-video.mp4" type="video/mp4">
</video>

<!-- HEADER: Logo + Main Nav (with Games dropdown) + Visitor Counter -->
<header>
<nav class="main-nav">
<li class="has-dropdown">
<a href="#games">Games</a> <!-- Shows dropdown on hover -->
<ul class="dropdown-menu"> ... </ul>
</li>
</nav>
<div class="visitor-count">Visitors: <span id="visitorCount"></span></div>
</header>

<!-- GAMES SECTION: Sub-nav filter + 6 categories of 4 cards each -->
<section id="games">
<!-- Sub-navigation: clicking a pill triggers showCategory() in JS -->
<div class="sub-nav"> ... </div>
<!-- Each category div: hidden/shown by JS filter -->
<div id="indoor" class="game-category"> ... </div>
<div id="outdoor" class="game-category"> ... </div>
</section>

<!-- TICKER: Fixed bottom bar with live date/time/location -->
<div class="ticker">
<div class="ticker-content" id="tickerContent"></div>
</div>

```

script.js — Annotated Key Functions

```
// ■■ VISITOR COUNTER ████████████████████████████████████████████  
  
// Reads count from localStorage, increments, saves back, displays  
let visits = localStorage.getItem("playfuloutings_visits") || 0;  
  
visits = Number(visits);  
  
if (isNaN(visits)) visits = 0; // Guard against corrupted data  
visits += 1;  
  
localStorage.setItem("playfuloutings_visits", visits);  
  
  
  
  
// ■■ GAME CATEGORY FILTER ████████████████████████████████████████████  
  
// showCategory: hides all .game-category divs, shows only selected
```



```
.game-card:hover { transform: translateY(-12px); }
```

```
/* YouTube lazy-load thumbnail wrapper (16:9 ratio) */
```

```
.yt-lazy { position: relative; padding-top: 56.25%;  
cursor: pointer; background: #000; }
```

```
.yt-lazy iframe { position: absolute; top: 0; left: 0;  
width: 100%; height: 100%; }
```

```
/* Ticker: fixed bottom scrolling bar */
```

```
.ticker { position: fixed; bottom: 0; z-index: 100;  
animation: ticker-scroll 58s linear infinite; }
```

9. User Guide

This guide explains how to use the PlayfulOutings website. No installation or account is required — simply open the website in any modern web browser.

Getting Started

Open the file **index.html** in any modern browser (Chrome, Firefox, Edge, or Safari). The site loads immediately with the background video and Indoor games displayed by default. Allow location access when prompted for the full ticker experience.

Navigating the Site

- Use the **top navigation bar** to jump to Home, Games, Itineraries, About Us, or Contact Us.
- Hover over **Games** in the nav bar to reveal a dropdown with all 6 game categories.
- Click any dropdown item to jump directly to that game category.
- Use the **sub-navigation pills** (Indoor, Outdoor, Kids, etc.) inside the Games section to filter games.
- The page scrolls smoothly to the selected section.

Browsing Games

- Each game card shows: **Materials Needed**, **Time Required**, and **How to Play** instructions.
- Click the **red play button** on any video thumbnail to watch a demo video for that game.
- Click "**Download Instructions (.docx)**" to download a Word document with full game details.
- Only the selected category's games are visible at one time — click another category pill to switch.

Contact & Location

- Scroll to the **Contact Us** section to find email, phone, and a Google Maps embed.
- If you allow location access, your coordinates will appear next to "Location:".
- Fill in your **Name** and **Message** in the contact form and click Send (demo mode).

Ticker & Visitor Count

- The **scrolling ticker** at the bottom shows the current date, time, your location, and fun messages.
- The **visitor counter** in the top-right corner tracks how many times the site has been opened on your device.

10. Developer's Guide

This guide is intended for developers who need to understand, maintain, or extend the PlayfulOutings codebase. The project is a fully client-side static website with no backend or build tools required.

Project File Structure

File / Folder	Description
index.html	Main HTML file — all sections and content
style.css	All styling — layout, components, animations, responsive
script.js	All JavaScript — filter, ticker, geolocation, lazy load
game_docs/	Folder containing 24 .docx game instruction files
videos/	Folder containing bg-video.mp4, bg-poster.png, Logo.png

Module Descriptions

Visitor Counter Module (script.js)

Uses `window.localStorage` to persist the visit count across sessions. On every page load, it reads "playfuloutings_visits", increments it by 1, writes it back, and updates the `#visitorCount` span in the header. NaN protection is included to handle corrupted storage.

Game Category Filter Module (script.js)

Implements `showCategory(id)`: queries all `.game-category` divs and sets `display:none` on all except the one whose id matches the argument. Updates the active class on `.sub-link` elements. Both `.sub-link` and `.dropdown-link` clicks invoke this function. On page load, all categories are hidden first, then `showCategory("indoor")` is called as the default.

YouTube Lazy Load Module (script.js)

Each game video is initially rendered as a thumbnail image with a red play button overlay (`.yt-lazy` div). On click, a YouTube iframe is created dynamically with `autoplay=1` in the src, replacing the thumbnail. A guard check prevents duplicate iframes if clicked twice.

Geolocation & Ticker Module (script.js)

Calls `navigator.geolocation.getCurrentPosition()` on load. On success, coordinates are displayed in the `#userLocation` span. The `updateTicker()` function assembles a string with current date, time, and location, sets it as `tickerContent.textContent` (doubled for seamless scroll loop), and is called every 1000ms via `setInterval`.

Smooth Scroll & Active Nav Module (script.js)

All anchor links are intercepted with `preventDefault()` and replaced with `scrollIntoView({behavior:"smooth"})`. A scroll event listener tracks the current visible section and applies the `.active` class to the matching nav link.

CSS Variables & Theming (style.css)

All colors are defined as CSS custom properties in :root. This means changing --primary updates button colors, card headers, labels, and borders simultaneously. The glassmorphism effect uses backdrop-filter:blur() with semi-transparent backgrounds.

Responsive Layout Module (style.css)

Three media query breakpoints: 992px (stacks header vertically), 768px (single-column game grid, vertical nav links, flex-row sub-nav), 480px (smaller fonts). Game grid uses CSS Grid with auto-fit and minmax(480px,1fr) on desktop.

Dropdown Menu Module (style.css + HTML)

The .dropdown-menu is positioned absolutely under the Games nav item. It uses opacity:0 and pointer-events:none by default, transitioning to opacity:1 on :hover and :focus-within of the parent li. JavaScript handles clicks to trigger showCategory() and smooth scroll.

How to Add a New Game

- 1. Add a new .game-card div inside the relevant .game-category div in index.html.
- 2. Include .info-item divs for Materials, Time, and How to Play.
- 3. Add a .yt-lazy div with the correct YouTube video ID as data-videoid.
- 4. Add a .download-btn anchor pointing to a new .docx file in the game_docs/ folder.
- 5. Create the corresponding .docx file with game details and place it in game_docs/.
- 6. No changes to CSS or JS are needed — the filter and lazy-load work automatically.

Browser Compatibility

Browser	Version Tested	Status
Google Chrome	120+	Fully Supported
Mozilla Firefox	121+	Fully Supported
Microsoft Edge	120+	Fully Supported
Safari	17+	Fully Supported
Opera	106+	Fully Supported